Listening to Seattle's birds with Convolutional Neural Networks

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Introduction

The identification of bird species through their calls plays a key role in biodiversity monitoring and conservation research [1]. This project explores whether convolutional neural networks (CNNs) can reliably classify bird species using spectrogram representations of their calls. By automating the identification process, the goal is to remove the need for expert knowledge and make high-volume bird species recognition accessible to a wider audience. Two classification problems were explored: a binary classification problem distinguishing House Sparrow and Song Sparrow bird calls, and a multiclass classification problem involving all twelve bird species in the dataset. The project aims to evaluate how different CNN architectures, preprocessing strategies, and data augmentation techniques affect classification performance.

Theoretical Background

Neural networks, inspired by the architecture of the human brain, consist of interconnected computational units called neurons, organized into layers. Each neuron receives input values, applies learned weights and biases, and passes the result through a nonlinear activation function. During training, the network iteratively adjusts these weights to minimize prediction error on the training data. For a single neuron, the output can be represented as:

$$z = \sum_{i=1}^{n} \mathrm{w}_1 \mathrm{x}_\mathrm{i} + \mathrm{b}$$

where x_i represents the input variables, w_i represents learned weights, and b is the bias term.

The neuron output is then transformed using an activation function, such as Rectified Linear Unit, Sigmoid or Softmax [2]. Neural networks are trained using backpropagation and gradient descent optimization. During each training iteration, predictions are compared against the true labels using a loss function. For multiclass classification, categorical cross-entropy loss is commonly used:

$$\mathrm{L} = \sum_{i=1}^{C} \mathrm{y}_\mathrm{i} \log(\hat{\mathrm{y}}_\mathrm{i})$$

where C is the number of classes, y_i is the true class indicator, and $\hat{y}_i$ is the predicted probability for class i [3]. The objective of training is to minimize this loss function through iterative updates to the model parameters.

Recurrent Neural Networks (RNNs) are another possible architecture for audio analysis because they are designed to model sequential and temporal dependencies [4]. However, RNNs are often computationally slower and more difficult to train on large spectrogram inputs. In contrast, convolutional neural networks (CNNs) are specifically designed to process spatially structured data such as images and spectrograms. They are particularly well suited for bird sound classification because they can automatically learn local acoustic patterns, including harmonics and chirps directly from spectrogram images. A CNN typically consists of convolutional layers, pooling layers, and fully connected dense layers. Convolutional layers apply small learnable filters across the input image to detect local features.Parameters that can be used to tune and improve CNN performance are convolutional layers, number of filters, kernel size, learning rate, batch size, dropout rate, and number of training epochs. Underfitting occurs when the network lacks sufficient capacity to learn meaningful patterns, while overfitting occurs when the model memorizes training data and performs poorly on unseen data. To improve generalization, several regularization techniques are commonly applied. Dropout randomly disables neurons during training to reduce memorization, while batch normalization stabilizes intermediate activations and improves gradient flow [5]. Model performance can be evaluated using multiple metrics beyond overall accuracy. Precision measures how often predicted bird labels are correct, recall measures how effectively the model identifies true bird recordings, and the F1-score balances precision and recall. Confusion matrices are also used to visualize which bird species were frequently confused with one another. Training and validation learning curves provide insight into overfitting and convergence behavior during optimization.

Methodology

The bird spectrogram dataset was loaded from an HDF5 file format using the h5py package in Python. The dataset consisted of 12 bird species: American Crow, American Robin, Bewick’s Wren, Black-capped Chickadee, Dark-eyed Junco, House Finch, House Sparrow, Northern Flicker, Red-winged Blackbird, Song Sparrow, Spotted Towhee, and White-crowned Sparrow. Each species contained spectrogram recordings with identical frequency and time dimensions of 128 and 517, respectively, while the number of recordings varied substantially between species, ranging from 37 to

630 samples. This uneven distribution of recordings introduces class imbalance, which is expected to be a key challenge during the training and evaluation of neural network models.

A spectrogram can be interpreted as a visual representation of sound over time. For example, a sample data tensor with dimensions (128, 517, 66) represents 66 separate bird recordings, where 128 corresponds to the number of frequency or pitch levels and 517 corresponds to the number of time slices. Lower frequency levels represent deeper sounds, while higher frequency levels correspond to sharper chirps and higher-pitched vocalizations. The spectrograms are stored on a logarithmic decibel (dB) scale, where values near 0 dB represent the loudest frequency regions. Audio below -80 dB is clipped because quieter signals are considered negligible background noise. Summary statistics from the spectrograms showed a minimum value of -80.0, maximum value of approximately 0.0, mean of -43.04, and standard deviation of 16.90. These statistics indicate that most regions of the spectrogram are relatively quiet and that bird vocalizations occupy only portions of the overall frequency space. The observed standard deviation also suggests meaningful variation and contrast within the spectrogram images, which is beneficial for convolutional neural network feature extraction.

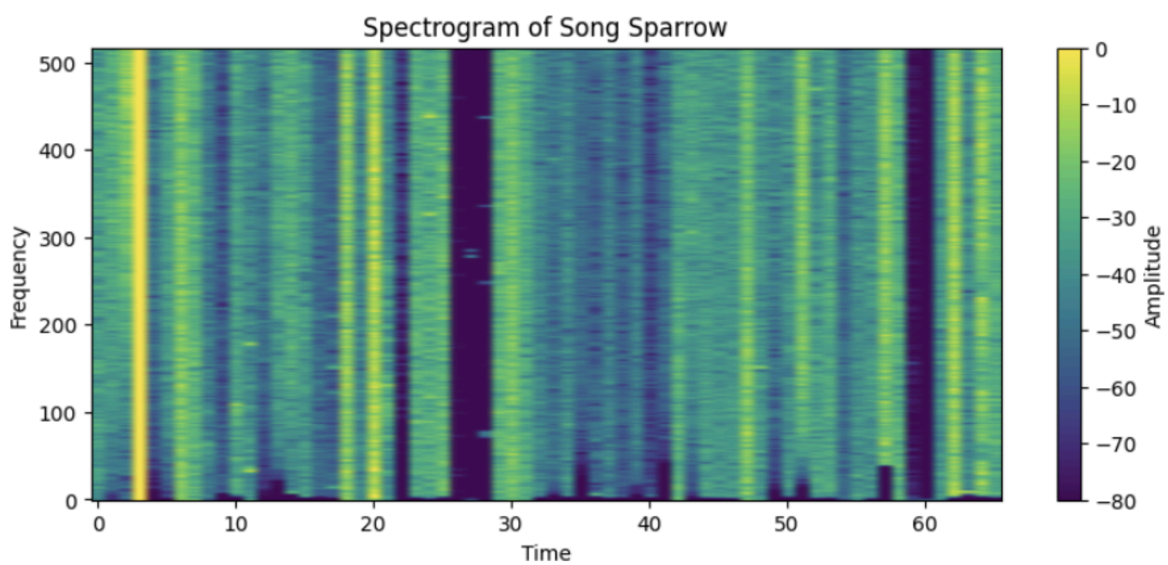

Figure 1. Sample spectrogram of Song Sparrow bird

To prepare the dataset for convolutional neural network modeling, a master dataset was first constructed by transposing the original tensors into the format expected by CNN architectures. The spectrogram values were then normalized to place all inputs on a consistent numerical scale. An additional channel dimension was also added because convolutional neural networks expect image-like input structures, even for grayscale spectrogram representations. From the master dataset, two task-specific subsets were created. The binary classification subset contained only House Sparrow and

Song Sparrow recordings. The multiclass subset retained all bird species while addressing class imbalance through a combination of controlled sampling and spectrogram augmentation. Initially, a target sample size of 300 recordings per species was established. Species with more than 300 recordings were randomly downsampled to reduce dominance by majority classes, while species with fewer than 300 recordings were supplemented using data augmentation techniques. Rather than duplicating recordings directly, spectrogram augmentation was applied to generate additional training examples for minority classes. Specifically, a SpecAugment approach was implemented by randomly masking small frequency bands and short temporal regions within the spectrograms. By generating augmented spectrograms only for minority classes, the dataset gains additional variability without artificially inflating already dominant classes.

Binary Classification Models

A binary convolutional neural network (CNN) classifier was developed to distinguish between House Sparrow and Song Sparrow spectrograms. The dataset was first divided into training and testing subsets using a stratified train-test split to preserve the original class proportions across both sets. Spectrogram tensors were normalized and formatted into grayscale image-like input for compatibility with convolutional neural network architectures. The baseline CNN architecture consisted of two convolutional layers with ReLU activation functions, each followed by a max-pooling layer for dimensionality reduction and feature extraction. The extracted feature maps were flattened and passed into a dense hidden layer before the final sigmoid output layer for binary classification. Hyperparameters were initially selected conservatively to establish a stable baseline model and included 32 and 64 convolutional filters, a kernel size of (3,3), batch size of 16, and 10 training epochs using the Adam optimizer and binary cross-entropy loss function. The baseline model achieved an overall test accuracy of 88%, with particularly strong performance on the House Sparrow class. Precision, recall, and F1-score metrics were evaluated because the dataset exhibited moderate class imbalance between the two bird species. While overall accuracy was high, macro-level metrics revealed weaker minority-class performance for Song Sparrow, particularly in recall and F1-score, indicating that the model occasionally failed to identify true Song Sparrow recordings.

	Predicted House Sparrow	Predicted Song Sparrow
Actual House Sparrow	117	9
Actual Song Sparrow	13	40

Figure 2. Confusion matrix of first binary classification model

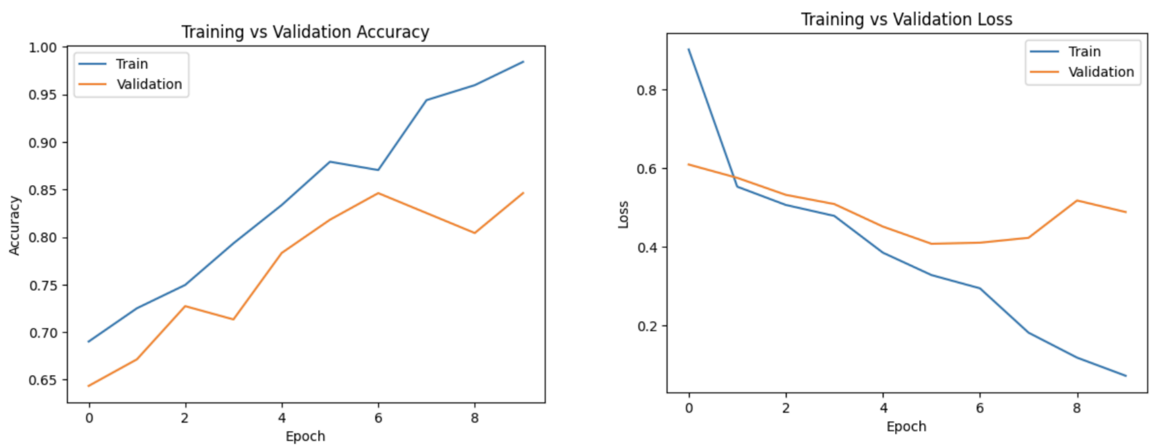

Figure 3. Training/Validation accuracy and loss for first binary classification model

A second CNN model was developed to investigate whether adding regularization and addressing imbalance could improve performance. The revised model kept the same overall CNN structure but introduced several modifications, including a dropout layer with a dropout rate of 0.5, balanced class weights during training, and early stopping based on validation loss. The dropout layer should reduce overfitting by randomly deactivating neurons during training, while class weighting would penalize classification errors on the minority class more heavily. Early stopping was added to halt training once validation performance plateaued. Multiple epoch settings were tested, with training extended up to 30 epochs while allowing early stopping to determine the optimal stopping point automatically. Despite these adjustments, the revised model produced lower overall performance, achieving 78% accuracy compared to the baseline model's 88%. Although Song Sparrow recall remained relatively stable (0.74 compared to 0.75 in the baseline), precision and F1-score decreased substantially, suggesting that the model became overly sensitive to minority-class predictions and introduced more false positives. Additionally, training accuracy approached 100% while validation accuracy plateaued near 83%, indicating persistent overfitting despite the introduction of regularization techniques. Comparisons between the two architectures suggest that the original CNN architecture already possessed sufficient capacity for the binary classification task and that other more aggressive measures may have reduced overall model stability.

	First Binary Model (Best)		**Second Binary Model**	
	House Sparrow	*Song Sparrow*	*House Sparrow*	*Song Sparrow*
Precision	0.90	0.82	0.88	0.60
Recall	0.93	0.75	0.79	0.74
F-1 Score	0.91	0.78	0.83	0.66
Accuracy	0.88		0.78	

Figure 4. Comparison of results for first and second binary classification models

Multiclass Classification Models

The first multiclass convolutional neural network was designed with a relatively simple baseline architecture. The model consisted of two convolutional layers with 32 and 64 filters respectively, each followed by max pooling layers for dimensionality reduction. Feature maps were flattened and passed through a dense hidden layer before the final softmax output layer for multiclass classification. As previously mentioned in the "Methodology" section, this architecture was used on a balanced multiclass subset with controlled sampling and SpecAugment spectrogram augmentation. Frequency masking and time masking were applied to minority classes to increase training diversity while preserving the overall acoustic structure of bird calls. The resulting model achieved an overall multiclass accuracy of 46% with macro-average precision, recall, and F1-scores of approximately 0.46–0.50. Several species achieved moderate classification performance, including Black-capped Chickadee, White-crowned Sparrow, and Northern Flicker, suggesting that these species possessed relatively distinctive acoustic patterns. However, species such as Bewick's Wren, House Sparrow, Song Sparrow, and Spotted Towhee continued to produce relatively poor F1-scores, indicating confusion between acoustically similar bird calls. The model also exhibited signs of overfitting, with training loss decreasing rapidly while validation loss increased sharply after only a few epochs. Training the model required approximately 920.12 seconds, reflecting the substantial computational cost associated with large spectrogram inputs even for relatively simple CNN architectures.

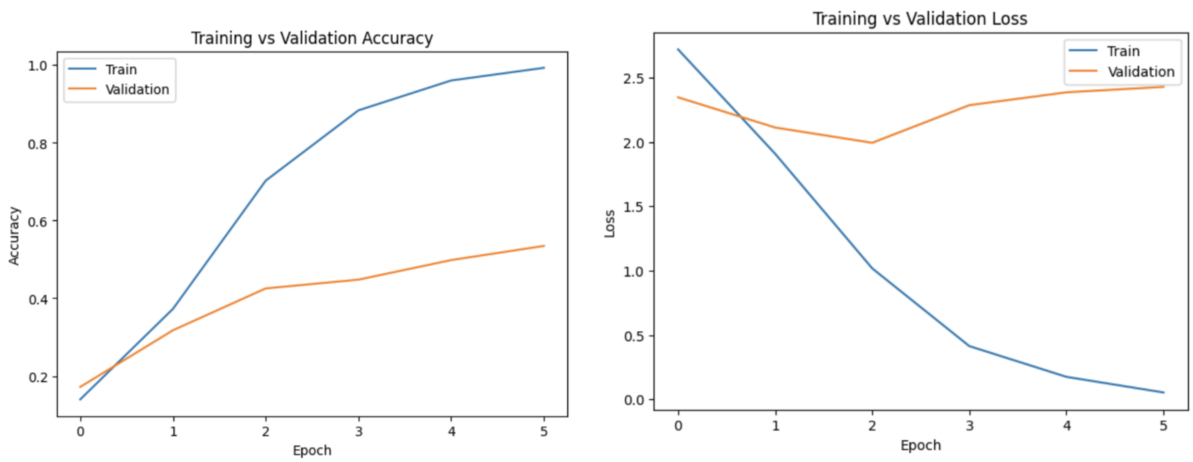

Figure 5. Training/Validation accuracy and loss for first multiclass classification model

The second multiclass convolutional neural network introduced several architectural improvements. This model added a third convolutional block with 128 filters, incorporated Batch Normalization after each convolutional layer, replaced the Flatten layer with Global Average Pooling, and introduced dropout regularization before the final classification layer. The learning rate was also reduced to improve optimization stability during training. These modifications substantially improved overall performance, increasing multiclass accuracy from 46% to 56% while also improving macro-average precision, recall, and F1-score metrics. Several bird species demonstrated notable gains in classification performance, particularly House Sparrow, Black-capped Chickadee, House Finch and Northern Flicker. Northern Flicker and White-crowned Sparrow achieved especially high recall values (0.90 and 0.87 respectively), indicating that the CNN was highly effective at identifying true recordings of these species. However, precision for these species remained lower, suggesting that the model occasionally over-predicted these classes when encountering acoustically similar bird calls. Species such as Spotted Towhee, Red-winged Blackbird, and Song Sparrow continued to demonstrate lower F1-scores, indicating that these vocalizations remained difficult for the CNN to distinguish reliably. Although the revised architecture substantially improved classification accuracy and generalization, these gains came at a major computational cost. Training time increased dramatically to approximately 12,463 seconds due to the deeper architecture, additional normalization operations, and increased feature extraction complexity.

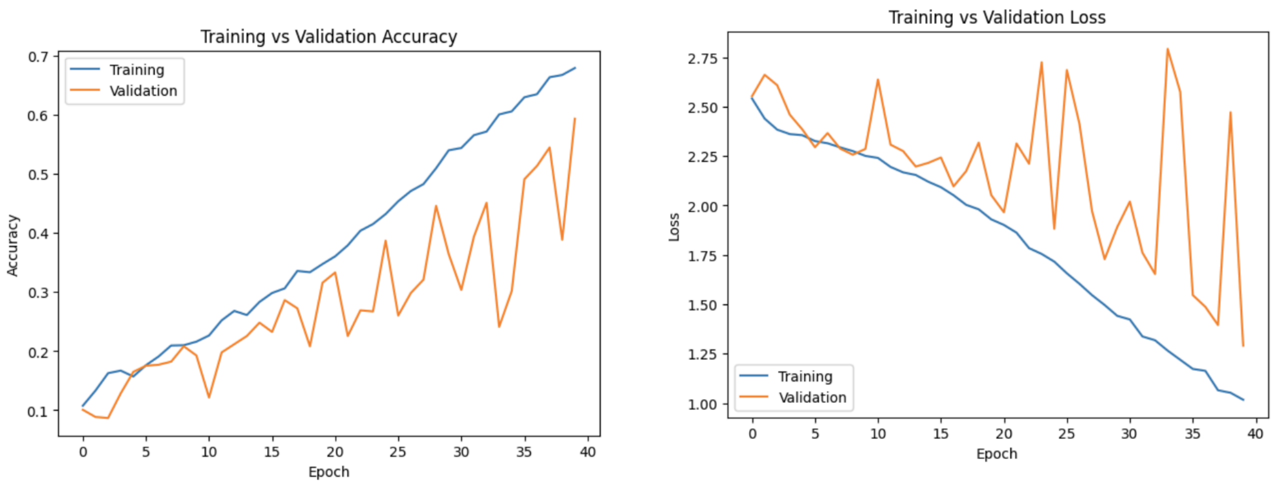

Figure 6. Training/Validation accuracy and loss for second multiclass classification model

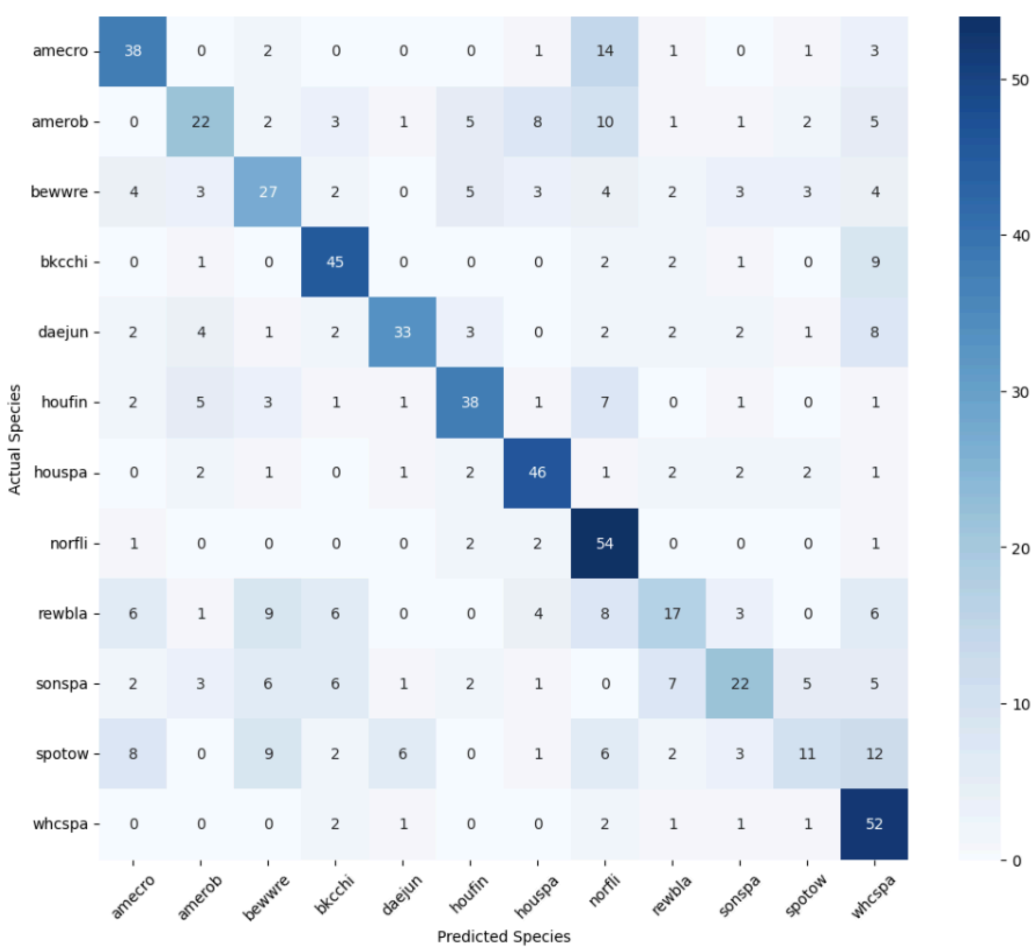

Figure 7. Confusion matrix of second multiclass classification model

Architecture	First Multiclass Model	Second Multiclass Model (Best)
Convolutional Layer	2	3
Batch Normalization	0	3
Max-pooling	2	3
Global average pooling	0	1
Dense	2	2
Dropout	0	1
Flatten	1	0
Early stopping	Yes	Yes
Accuracy	0.4556	0.5625

Figure 8. Architecture comparison of first and second multiclass classification model

	Precision		**Recall**		**F-1 Score**	
	1st Model	*2nd Model*	*1st Model*	*2nd Model*	*1st Model*	*2nd Model*
American Crow	0.55	0.60	0.48	0.63	0.51	0.62
American Robin	0.59	0.54	0.48	0.37	0.53	0.44
Bewick's Wren	**0.27**	0.45	0.37	0.45	**0.31**	0.45
Black-capped Chickadee	0.46	0.65	0.70	0.75	0.55	0.70
Dark-eyed Junco	0.40	**0.75**	0.38	0.55	0.39	0.63
House Finch	0.76	0.67	0.42	0.63	0.54	0.65
House Sparrow	0.49	0.69	0.30	0.77	0.37	**0.72**

Northern Flicker	**0.79**	0.49	0.45	**0.90**	**0.57**	0.64
Red-winged Blackbird	0.43	0.46	0.48	0.28	0.46	0.35
Song Sparrow	0.53	0.56	**0.27**	0.37	0.36	0.44
Spotted Towhee	0.32	**0.42**	0.42	**0.18**	0.36	**0.26**
White-crowned Sparrow	0.39	0.49	**0.72**	0.87	0.51	0.62

Figure 9. Precision, recall, F-1 score of all bird species in both multiclass classification model

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Test Audios

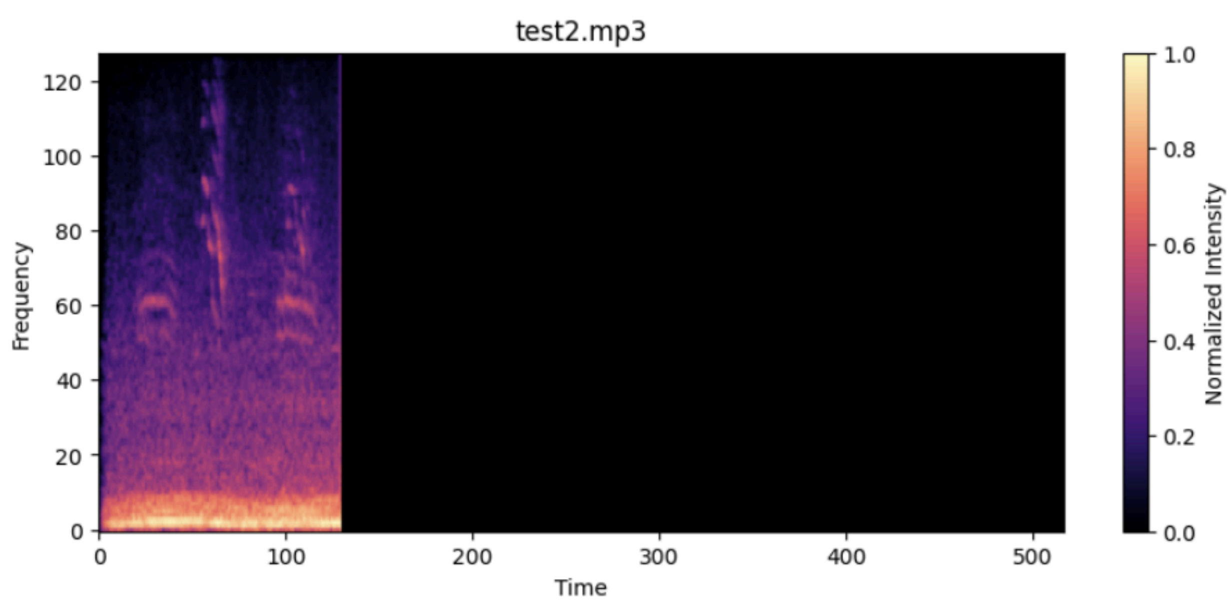

Figure 10. Spectrogram of the second test audio

After developing and evaluating the multiclass convolutional neural network architectures, the final improved model was applied to three previously unseen audio clips. Because the neural network was trained on spectrogram inputs rather than raw audio waveforms, each test clip first underwent the same preprocessing pipeline used during model training to ensure consistency between training and inference data distributions. The audio clips were loaded using the librosa package and resampled to

22050 Hz to match the preprocessing specifications of the original dataset. The first three seconds of each clip were extracted to maintain consistency with the training spectrogram duration. A Mel spectrogram with 128 frequency bins was then generated from each audio recording. The power spectrograms were then converted into decibel-scaled spectrograms using logarithmic scaling, producing the same type of input representation used during CNN training. Following spectrogram generation, each spectrogram was normalized to a 0 to 1 range. Since the time dimension of newly generated spectrograms may differ slightly depending on the exact clip duration and audio structure, the spectrograms were padded or truncated to the fixed dimensions of 128 × 517 used throughout model training. Finally, a single grayscale channel dimension was added so the input tensor matched the expected CNN input shape.

The processed spectrograms were then passed through the trained CNN using the model's softmax output layer to generate probability distributions across all twelve bird species. The species with the highest probability was selected as the primary predicted label for each audio clip. All three test clips were ultimately classified as White-crowned Sparrow. However, further analysis of the probability distributions, spectrogram visualizations, and audio playback suggested that some clips may contain multiple simultaneously singing bird species. Although the CNN was trained as a single-label multiclass classifier and therefore predicts only one dominant class per recording, the secondary probability outputs still provide useful exploratory information regarding potential overlapping bird calls. For the second audio, White-crowned Sparrow remained the dominant prediction, but Spotted Towhee also received a relatively elevated probability score (~0.13) compared to most other classes. Visual inspection of the spectrogram revealed two distinct temporal-frequency call structures: one consisting of shorter, sharper chirps and another containing longer singing patterns. Listening to the audio clip similarly suggested overlapping bird calls with differing acoustic characteristics. This combination of spectrogram structure and secondary probability evidence suggests the possible presence of a Spotted Towhee vocalization alongside the dominant White-crowned Sparrow call. Similarly, the third audio appeared to contain both White-crowned Sparrow and Black-capped Chickadee singing. The White-crowned Sparrow vocalizations appeared as higher-frequency chirp structures, while the suspected Black-capped Chickadee calls exhibited lower-frequency and deeper tonal characteristics.

	Top Prediction	Probability	Second Prediction	Probability	Possible Multi-Bird?
test1	White-crowned Sparrow	0.1909	House Sparrow	0.1547	No
test2	White-crowned Sparrow	0.1331	Spotted Towhee	0.1327	Yes
test3	White-crowned Sparrow	0.1858	Black-capped Chickadee	0.1832	Yes

Figure 11. Results of test audios

Discussion

One of the main difficulties is the highly imbalanced dataset. While some bird species contained more than 600 recordings, other species contained fewer than 40 samples. To address this issue, controlled downsampling and SpecAugment-inspired spectrogram augmentation were implemented to artificially increase variability within minority classes. Although these augmentation techniques substantially improved multiclass performance, dataset size and diversity remained important limitations throughout the project. Computational cost also became a major consideration during model development. The initial multiclass CNN architecture required approximately 920 seconds of training time, while the deeper and more sophisticated multiclass architecture required over 12,462 seconds to train. The dramatic increase in computational time reflected the significantly larger feature extraction complexity associated with additional convolutional layers, batch normalization, global average pooling, and increased parameter learning. These results highlight an important tradeoff in deep learning applications between model complexity, computational expense, and classification performance. Several bird species proved substantially more difficult for the CNN to classify accurately than others. Species such as Spotted Towhee, Song Sparrow, Red-winged Blackbird, and Bewick's Wren consistently achieved lower F1-scores across multiclass experiments. In contrast, species such as House Sparrow, Black-capped Chickadee, Northern Flicker, and White-crowned Sparrow demonstrated relatively strong classification performance.

Although convolutional neural networks were ultimately selected as the primary modeling approach, several alternative machine learning architectures could also have been explored. Traditional machine learning models such as Support Vector Machines (SVMs), Random Forests, or k-Nearest Neighbors could potentially classify handcrafted acoustic features extracted from the spectrograms. Recurrent Neural Networks (RNNs) could also be applied because bird calls contain temporal sequential information. Transfer learning approaches using pretrained image or audio architectures such as EfficientNet, ResNet, MobileNet, or BirdNET could also substantially improve classification accuracy, particularly when training data is limited [6].

Neural networks nevertheless represent a highly appropriate choice for this application because bird spectrograms contain complex nonlinear spatial and temporal patterns that are difficult to manually engineer using traditional statistical methods. CNNs are especially effective because they automatically learn hierarchical acoustic representations directly from spectrogram images without requiring handcrafted feature extraction. The ability of CNNs to identify localized spectro-temporal structures such as harmonics, chirps, and pitch contours makes them particularly well suited for bird sound classification tasks.

Conclusion

This project investigated the use of convolutional neural networks for automated bird species classification using spectrogram representations of bird calls. Both binary and multiclass classification problems were explored in order to evaluate the effectiveness of CNN-based feature learning. The binary classification task involving House Sparrow and Song Sparrow achieved strong overall performance, demonstrating that relatively simple CNN architectures can effectively distinguish between two bird species. The multiclass problem proved substantially more challenging due to the increased number of species, class imbalance, and acoustic similarities between several bird calls. The project also highlighted important tradeoffs between model complexity and computational cost.

Evaluation on three previously unseen audio clips demonstrated that the final CNN architecture generalized reasonably well to external recordings. All three clips were predicted as White-crowned Sparrow, while secondary probability outputs, spectrogram inspection, and auditory analysis suggested the likely presence of additional bird species within some recordings.

Automated bird sound recognition systems have significant potential applications in ecological monitoring, conservation biology, biodiversity assessment, environmental surveying, and large-scale wildlife observation. Deep learning models capable of identifying bird species from passive environmental recordings could dramatically reduce the labor required for ecological field studies and improve the scalability of biodiversity monitoring programs.

Future work could further improve performance through the use of larger and more diverse datasets, advanced audio augmentation techniques, multi-label classification architectures capable of detecting simultaneous bird species, and transfer learning approaches using pretrained audio or image-based neural networks such as BirdNET, EfficientNet, or ResNet.

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